

## BPHO ROUND 0 · QUESTION PAPER

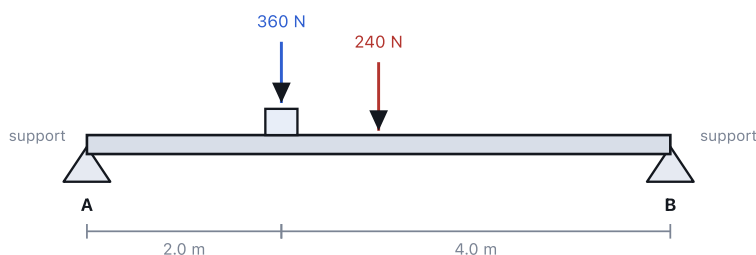
## Drill bank section 3

25 questions · 60 minutes · single answer, 5 options · **no calculator** · no negative marking*Original questions written for this course, in the Round 0 format. Not BPhO material.***Before you start**

- Answer every question. There is no negative marking, so a blank is a wasted mark.
- No calculator. Every answer here is reachable by hand, and the numbers are chosen to divide cleanly.
- Work on paper, then mark yourself against the markscheme PDF.
- The 2025 UK qualifying line was 11 out of 25, so about 44% is the target to beat.

**Q1** · A loaded beam on two supports: the reaction at one end **C**

A uniform beam **AB** is 6.0 m long and weighs 240 N. It rests on a support at **A** and a support at **B**. A load of 360 N is placed on the beam 2.0 m from **A**, as shown.



What is the upward force exerted by the support at **A**?

- |                |                |
|----------------|----------------|
| <b>A</b> 300 N | <b>B</b> 240 N |
| <b>C</b> 360 N | <b>D</b> 480 N |
| <b>E</b> 180 N |                |

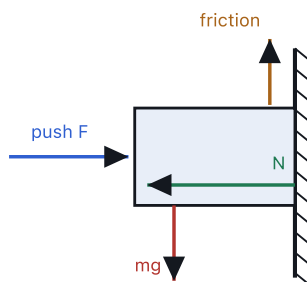
**Q2** · Dimensional analysis: the vibration frequency of a liquid droplet **A**

A small droplet of liquid, left to itself, vibrates about its spherical shape. The frequency **f** of that vibration depends only on the surface tension  **$\gamma$**  of the liquid, its density  **$\rho$** , and the radius  **$r$**  of the droplet. Which expression is dimensionally a frequency?

- |                                     |                                       |
|-------------------------------------|---------------------------------------|
| <b>A</b> $\sqrt{(\gamma/(\rho r))}$ | <b>B</b> $\sqrt{(\gamma r^3/\rho)}$   |
| <b>C</b> $\sqrt{(\gamma \rho/r^3)}$ | <b>D</b> $\sqrt{(\gamma/(\rho r^3))}$ |
| <b>E</b> $\gamma/(\rho r^3)$        |                                       |

**Q3** · Friction against a vertical wall: the smallest push that holds a block up **C**

A block of mass 2.0 kg is held against a rough vertical wall by a horizontal push **F**, as shown. The coefficient of friction between the block and the wall is 0.40. The block is on the point of sliding down the wall.

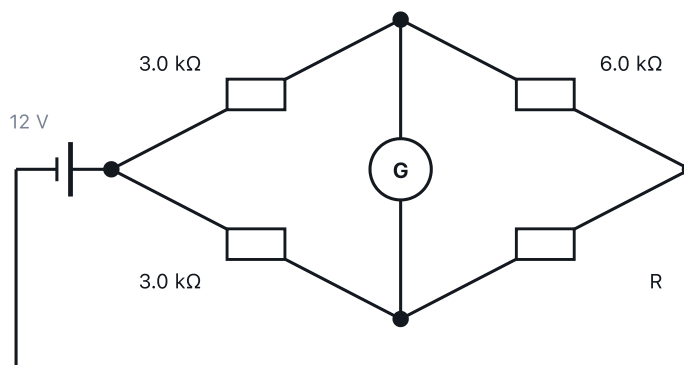


What is the smallest value of **F** that can hold the block in place?

- |                |               |
|----------------|---------------|
| <b>A</b> 8.0 N | <b>B</b> 50 N |
| <b>C</b> 20 N  | <b>D</b> 80 N |
| <b>E</b> 5.0 N |               |
- 

**Q4** · A balanced bridge: the fourth resistance **H**

The bridge below balances: no current flows through the detector **G** when **R** has one particular value.



What is that value of **R**?

- |                 |                 |
|-----------------|-----------------|
| <b>A</b> 6.0 kΩ | <b>B</b> 2.0 kΩ |
| <b>C</b> 18 kΩ  | <b>D</b> 4.5 kΩ |
| <b>E</b> 3.0 kΩ |                 |
-

**Q5** · Estimation: the power needed to climb a 10% hill on a bicycle **A**

A cyclist and bicycle together have a mass of about 80 kg. The cyclist climbs a hill whose gradient is 10% at a steady 5.0 m/s.

Which value is the closest estimate of the useful power the cyclist is delivering against gravity?

- |                |                 |
|----------------|-----------------|
| <b>A</b> 40 W  | <b>B</b> 2000 W |
| <b>C</b> 200 W | <b>D</b> 4000 W |
| <b>E</b> 400 W |                 |
- 

**Q6** · A three-stage journey: the average speed is a time-weighted mean, not a simple one **B**

A train starts from rest and accelerates uniformly at  $0.50 \text{ m s}^{-2}$  for 20 s. It then runs at constant speed for 30 s, and finally decelerates uniformly at  $1.0 \text{ m s}^{-2}$  until it comes to rest.

What is the average speed for the whole journey?

- |                                 |                                 |
|---------------------------------|---------------------------------|
| <b>A</b> $7.5 \text{ m s}^{-1}$ | <b>B</b> $10 \text{ m s}^{-1}$  |
| <b>C</b> $8.3 \text{ m s}^{-1}$ | <b>D</b> $9.0 \text{ m s}^{-1}$ |
| <b>E</b> $5.0 \text{ m s}^{-1}$ |                                 |
- 

**Q7** · Elastic collision: the speed of the heavier target **C**

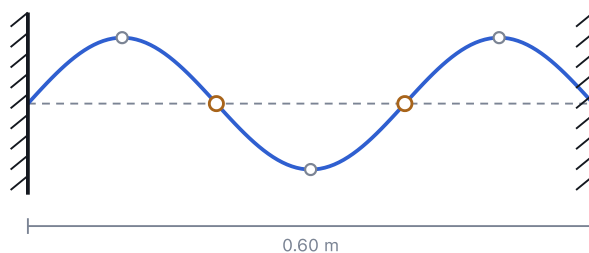
A particle of mass **m** moves at speed **u** along a smooth horizontal track. It collides head-on with a stationary particle of mass **3m**. The collision is perfectly elastic.

What is the speed of the particle of mass **3m** immediately after the collision?

- |                      |                     |
|----------------------|---------------------|
| <b>A</b> <b>u/2</b>  | <b>B</b> <b>u/3</b> |
| <b>C</b> <b>u/4</b>  | <b>D</b> <b>u</b>   |
| <b>E</b> <b>3u/4</b> |                     |
-

**Q8** · A string fixed at both ends: the third harmonic and its frequency **F**

A string of length 0.60 m is fixed at both ends and vibrates in the pattern shown, which is its third harmonic. The speed of transverse waves on the string is 120 m/s.



What is the frequency of this vibration?

- |                 |                 |
|-----------------|-----------------|
| <b>A</b> 100 Hz | <b>B</b> 300 Hz |
| <b>C</b> 200 Hz | <b>D</b> 600 Hz |
| <b>E</b> 900 Hz |                 |
- 

**Q9** · Apparent depth: how far below the surface a submerged object seems to be **G**

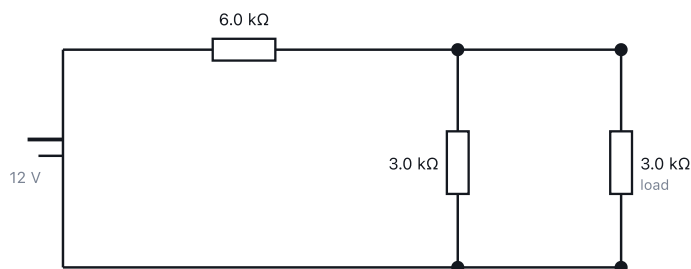
A stone lies on the bottom of a pool of water 2.4 m deep. The refractive index of water is 1.5. An observer looking straight down from directly above sees the stone.

At what depth below the surface does the stone appear to be?

- |                 |                |
|-----------------|----------------|
| <b>A</b> 3.6 m  | <b>B</b> 2.4 m |
| <b>C</b> 0.80 m | <b>D</b> 1.2 m |
| <b>E</b> 1.6 m  |                |
-

**Q10** · A loaded potential divider: the voltage actually delivered H

The circuit below shows a 12 V supply connected to a 6.0 kΩ resistor in series with a 3.0 kΩ resistor. A 3.0 kΩ load is connected in parallel with the 3.0 kΩ resistor.



What is the potential difference across the load?

- |                |                |
|----------------|----------------|
| <b>A</b> 4.0 V | <b>B</b> 4.8 V |
| <b>C</b> 3.0 V | <b>D</b> 2.4 V |
| <b>E</b> 1.2 V |                |

**Q11** · Ice at  $-10\text{ }^{\circ}\text{C}$  into water at  $30\text{ }^{\circ}\text{C}$ : whether all the ice can melt, and what is left if it cannot J

0.20 kg of ice at  $-10\text{ }^{\circ}\text{C}$  is added to 0.40 kg of water at  $30\text{ }^{\circ}\text{C}$  in an insulated container. The specific heat capacity of ice is  $2100\text{ J kg}^{-1}\text{ K}^{-1}$ , of water is  $4200\text{ J kg}^{-1}\text{ K}^{-1}$ , and the specific latent heat of fusion of ice is  $3.3 \times 10^5\text{ J kg}^{-1}$ .

What is the final state of the mixture?

- |                                                                          |                                                                    |
|--------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>A</b> At $0\text{ }^{\circ}\text{C}$ , with 0.060 kg of ice remaining | <b>B</b> At $0\text{ }^{\circ}\text{C}$ , with all the ice melted  |
| <b>C</b> At $0\text{ }^{\circ}\text{C}$ , with 0.14 kg of ice remaining  | <b>D</b> At $22\text{ }^{\circ}\text{C}$ , with all the ice melted |
| <b>E</b> At $0\text{ }^{\circ}\text{C}$ , with none of the ice melted    |                                                                    |

**Q12** · Nuclear density: why it is the same in every nucleus K

The radius of a nucleus is given by  $r = r_0 A^{1/3}$ , where  $A$  is the nucleon number and  $r_0$  is a constant.

How does the density of nuclear matter change when  $A$  increases from 27 to 216?

- |                                        |                                        |
|----------------------------------------|----------------------------------------|
| <b>A</b> It is unchanged               | <b>B</b> It increases by a factor of 8 |
| <b>C</b> It decreases by a factor of 8 | <b>D</b> It increases by a factor of 2 |
| <b>E</b> It decreases by a factor of 2 |                                        |

**Q13** · Scaling: by what power of  $N$  the total surface area grows when a cube is cut into  $N$  cubes **A**

A solid cube is cut up into  $N$  identical smaller cubes, where  $N$  is a perfect cube.

By what factor does the total surface area of all the pieces together exceed the surface area of the original cube?

**A**  $N^{1/3}$

**B**  $N^{2/3}$

**C**  $N$

**D**  $N^{1/2}$

**E**  $N^3$

**Q14** · A particle whose deceleration is proportional to its speed: the distance to rest **B**

A particle moves along a straight line with speed  $v$ . Throughout its motion its acceleration is  $a = -kv$ , where  $k$  is a positive constant.

The particle has speed  $u$  at  $t = 0$ . How far does it travel before coming to rest?

**A**  $uk$

**B**  $u/k$

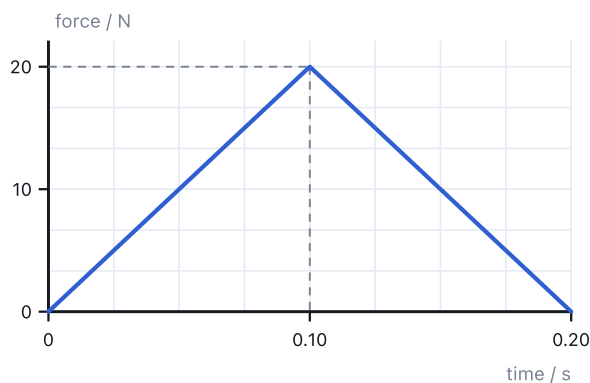
**C**  $u/(2k)$

**D**  $2u/k$

**E**  $u/(k \ln 2)$

**Q15** · Impulse from a force-time graph, and the speed it gives a ball **C**

A ball of mass  $2.0 \text{ kg}$  is at rest on a smooth horizontal surface. A variable force acts on it for  $0.20 \text{ s}$ ; the graph shows how the force varies with time.



What is the speed of the ball immediately after the force has acted?

**A**  $2.0 \text{ m/s}$

**B**  $0.50 \text{ m/s}$

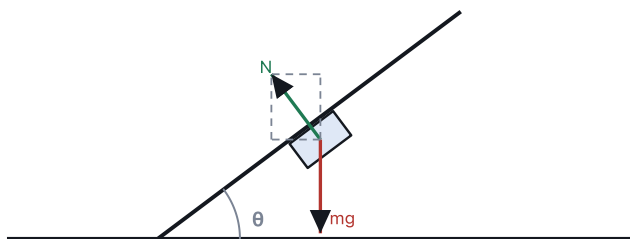
**C**  $4.0 \text{ m/s}$

**D**  $1.0 \text{ m/s}$

**E**  $20 \text{ m/s}$

**Q16** · A banked track: the angle at which no friction is needed D

A circular track of radius 30 m is banked at an angle  $\theta$  to the horizontal, as shown. A car travels round it at 15 m/s without any reliance on friction. Take  $g = 10 \text{ m/s}^2$ .



cross-section of the track, looking along the road

What is  $\theta$ ?

- |                     |                     |
|---------------------|---------------------|
| <b>A</b> $37^\circ$ | <b>B</b> $53^\circ$ |
| <b>C</b> $45^\circ$ | <b>D</b> $49^\circ$ |
| <b>E</b> $27^\circ$ |                     |

**Q17** · Sizing a cable from a breaking stress: the diameter as an algebraic expression E

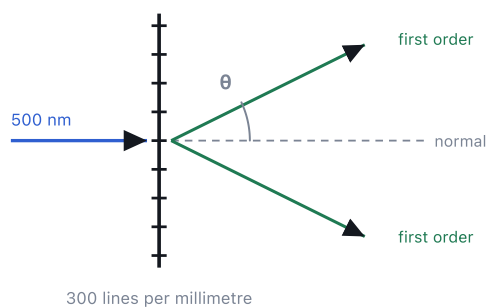
A steel cable is to support a load of mass  $m$ . The breaking stress of the steel is  $\sigma$  and the gravitational field strength is  $g$ .

Which expression gives the smallest diameter the cable may have?

- |                                       |                                      |
|---------------------------------------|--------------------------------------|
| <b>A</b> $\sqrt{(4mg / (\pi\sigma))}$ | <b>B</b> $\sqrt{(mg / (\pi\sigma))}$ |
| <b>C</b> $4mg / (\pi\sigma)$          | <b>D</b> $\sqrt{(4\pi mg / \sigma)}$ |
| <b>E</b> $\sqrt{(4mg\sigma / \pi)}$   |                                      |

**Q18** · The highest diffraction order a grating can produce **F**

A diffraction grating has 300 lines per millimetre. Monochromatic light of wavelength 500 nm falls on it normally, as shown.

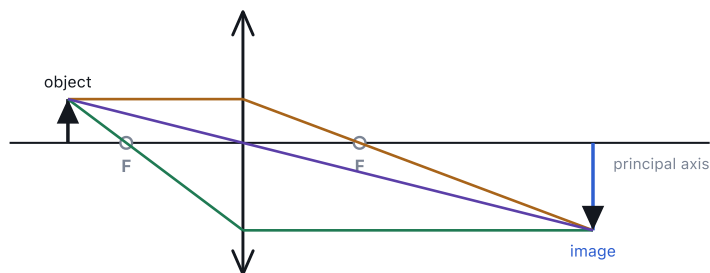


What is the highest order of maximum that can be observed?

- |             |             |
|-------------|-------------|
| <b>A</b> 7  | <b>B</b> 12 |
| <b>C</b> 13 | <b>D</b> 1  |
| <b>E</b> 6  |             |

**Q19** · A converging lens: the linear magnification **G**

An object is placed 30 cm from a thin converging lens of focal length 20 cm. The ray diagram shows the construction.



What is the linear magnification produced by the lens?

- |              |               |
|--------------|---------------|
| <b>A</b> 3   | <b>B</b> 2    |
| <b>C</b> 1.5 | <b>D</b> 0.67 |
| <b>E</b> 0.5 |               |



**Q20** · Two lamps of different power rating in series: which is brighter H

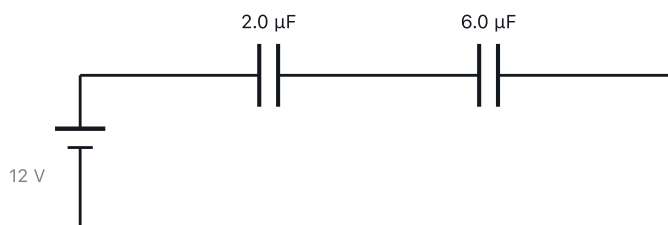
Two filament lamps are marked 60 W and 100 W, both rated for use on 240 V. They are connected in series across a 240 V supply.

Which lamp is the brighter, and why?

- A** The 100 W lamp, because it is rated at the higher power
- B** Neither, because the same current through both makes the powers equal
- C** The 100 W lamp, because its lower resistance lets more current through it
- D** The 60 W lamp, because it has the greater resistance and the current through both is the same
- E** Neither lights, because the two resistances in series are too large for a 240 V supply

**Q21** · Capacitors in series: how the supply divides between them I

A 2.0  $\mu\text{F}$  capacitor and a 6.0  $\mu\text{F}$  capacitor are connected in series across a 12 V supply, as shown.



What is the potential difference across the 2.0  $\mu\text{F}$  capacitor?

- A** 9.0 V
- B** 3.0 V
- C** 6.0 V
- D** 4.0 V
- E** 12 V

**Q22** · Linear thermal expansion: how much longer a heated rod becomes J

A brass rod is 2.0 m long at 20 °C. The coefficient of linear expansion of brass is  $1.9 \times 10^{-5}$  per °C.

How much longer is the rod at 120 °C?

- A** 7.6 mm
- B** 4.6 mm
- C** 3.8 mm
- D** 1.9 mm
- E** 0.38 mm

**Q23** · Mass-energy equivalence: the energy released by a mass defect K

In a nuclear reaction the total mass of the products is less than the total mass of the reactants by  $3.6 \times 10^{-30} \text{ kg}$ . Take the speed of light as  $3.0 \times 10^8 \text{ m/s}$ .

How much energy is released?

- |                                          |                                          |
|------------------------------------------|------------------------------------------|
| <b>A</b> $1.1 \times 10^{-21} \text{ J}$ | <b>B</b> $3.2 \times 10^{-13} \text{ J}$ |
| <b>C</b> $4.0 \times 10^{-47} \text{ J}$ | <b>D</b> $2.9 \times 10^4 \text{ J}$     |
| <b>E</b> $3.2 \times 10^{-14} \text{ J}$ |                                          |
- 

**Q24** · The photoelectric effect: the stopping potential for a given wavelength L

Light of wavelength 400 nm falls on a metal surface whose work function is 2.0 eV.

What is the stopping potential for the emitted photoelectrons?

- |                |                 |
|----------------|-----------------|
| <b>A</b> 3.1 V | <b>B</b> 5.1 V  |
| <b>C</b> 2.0 V | <b>D</b> 0.55 V |
| <b>E</b> 1.1 V |                 |
- 

**Q25** · Upthrust: what a spring balance reads with the load under water M

A stone has a volume of  $2.0 \times 10^{-4} \text{ m}^3$  and a mass of 0.50 kg. It is lowered until it is completely under water and weighed on a spring balance. The density of water is  $1000 \text{ kg/m}^3$  and  $g = 10 \text{ m/s}^2$ .

What does the balance read?

- |                |                |
|----------------|----------------|
| <b>A</b> 5.0 N | <b>B</b> 2.0 N |
| <b>C</b> 7.0 N | <b>D</b> 3.0 N |
| <b>E</b> 0 N   |                |